

IndiaKart E-Commerce Analytics Project

Project Overview

IndiaKart E-Commerce Analytics is an end-to-end data analytics project that demonstrates the complete analytics lifecycle from raw data to executive dashboards. The project uses Python, SQL, PostgreSQL, Excel, and Power BI to clean, analyze, visualize, and present business insights for an e-commerce business.

Business Objectives

- Analyze sales and revenue trends.
- Understand customer purchasing behavior.
- Evaluate product and category performance.
- Monitor inventory and supplier efficiency.
- Analyze payments and product returns.
- Support business decisions using interactive dashboards.

Technology Stack

Category	Tools
Programming	Python, Pandas, NumPy
Database	PostgreSQL, SQL
Visualization	Power BI, Excel, Matplotlib
Version Control	GitHub

Project Workflow

Raw Data → Data Cleaning → PostgreSQL Database → SQL Analysis → Python EDA → KPI Development → Power BI Dashboard → Business Recommendations

Key KPIs

- Total Revenue
- Net Revenue
- Average Order Value (AOV)
- Customer Lifetime Value (CLV)
- Return Rate
- Payment Success Rate
- Inventory Value

Business Recommendations

- Improve inventory planning for fast-moving products.
- Strengthen loyalty programs for repeat customers.
- Reduce return rates through quality control.
- Monitor supplier performance using KPIs.
- Use dashboards for data-driven decision-making.

Conclusion

This project demonstrates practical skills in Python, SQL, PostgreSQL, Power BI, data visualization, KPI development, and business storytelling, making it a strong portfolio project for Data Analyst roles.